Temporal-Feedback Self-Training for Semi-Supervised Object Detection in Remote Sensing Images

Xiaoqian Zhu, Xiangrong Zhang, *Senior Member, IEEE*, Tianyang Zhang, Xu Tang, *Senior Member, IEEE*, Puhua Chen, *Senior Member, IEEE*, Huiyu Zhou, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Although modern remote sensing object detection (RSOD) methods have achieved advanced performance, they heavily rely on a large amount of annotated data. This article explores semi-supervised RSOD to mitigate annotation costs, leveraging recent extensive research in generic semi-supervised object detection (SSOD) based on the self-training paradigm. Current SSOD methods encounter challenges in adapting to remote sensing images (RSIs) due to the complexity and variability of RSIs. Two key issues remain underexplored: the noise in pseudo-labels caused by model instability and the difficulty in distinguishing similar categories. This article introduces the temporal-feedback self-training (TST) framework, a novel approach to tackle these challenges in semi-supervised RSOD. TST consists of two components: temporal consistency-based pseudo-labels certainty estimation (TCE) and temporal self-feedback feature refinement (TSF). TCE addresses pseudo-label noise during training by evaluating the stability of pseudo-label classification and localization over time series to assess the quality of pseudo-labels. On the other hand, TSF enhances pseudo-label quality by dynamically identifying the model's confusing categories as feedback for feature refinement. Both components facilitate the progression of the self-training-based RSOD during training. We conducted extensive experiments on two challenging public datasets: DOTA and DIOR. The results demonstrate that the proposed TST and TCE components significantly improve the baseline model's performance, surpassing the state-of-the-art generic SSOD method. This suggests that our approach is more effective than generic SSOD methods in addressing the challenges posed by RSIs.**

***Index Terms*— Pseudo-labeling, remote sensing images (RSIs), self-training, semi-supervised object detection (SSOD).**

I. Introduction

REMOTE sensing object detection (RSOD) is one of the most fundamental and challenging tasks in the remote sensing field, which is widely applied in crop monitoring, resource exploration, traffic planning, urban construction, environmental monitoring, military reconnaissance, and other fields [1], [2], [3], [4]. In recent years, driven by the rapid advancement of deep learning, RSOD has achieved breakthrough performance. However, deep learning-based object detection heavily relies on the availability of a massive amount of annotated data. The density of objects in remote sensing images (RSIs), along with the presence of numerous small objects, makes accurate bounding box annotation for RSIs a time-consuming and labor-intensive task.

Semi-supervised learning (SSL) has garnered significant interest in the remote sensing field due to its remarkable potential to alleviate label dependency. Although semi-supervised RSOD remains relatively unexplored, there has been substantial progress in the development of generic semi-supervised object detection (SSOD). Mainstream SSOD methods are based on the self-training paradigm [5], [6], [7], [8], which involves mining large amounts of unlabeled data by predicting pseudo-labels on them while simultaneously training the model on labeled data. However, the SSOD methods based on this paradigm heavily rely on the quality of the pseudo-labels. Earlier SSOD methods [9], [10], [11], [12], [13] use a high confidence threshold to filter out noisy pseudo-labels. Nevertheless, due to the multitask nature of object detection, high classification scores of pseudo-labels do not necessarily indicate high-quality localization. To fill this gap, some follow-up work has explored the use of box jitter [14], regression uncertainty prediction [15], and localization score [16], [17] as alternative ways to assess the localization quality of pseudo-labels in SSOD.

However, the complexity and variability of the background in RSIs intensify the noise in pseudo-labels, making it challenging for general SSOD methods to adapt to RSIs. These methods overlook another essential issue of SSOD with self-training patterns: the pseudo-label noise arising from model instability, which is particularly severe in RSIs with complex backgrounds. Self-training-based SSOD methods require generating pseudo-labels for unlabeled data while training the model itself. The model is unstable during the training, resulting in dramatic fluctuations of the pseudo-labels in the different training phases [as shown in Fig. 1(a)]. The instability of the pseudo-labels makes the model confused and severely degrades the detection performance. Moreover,

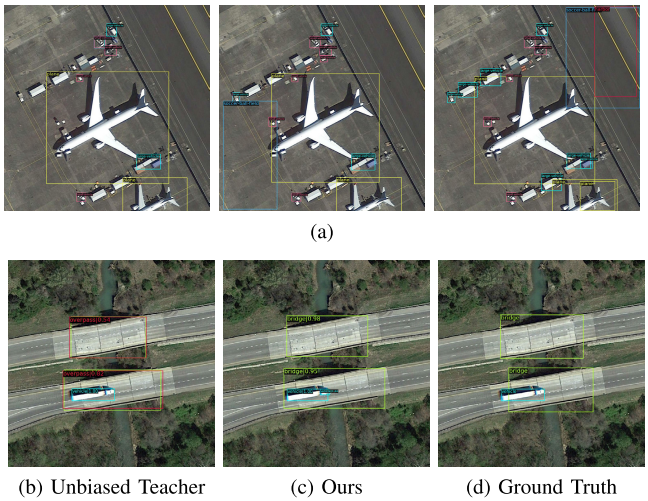
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Xiaoqian Zhu, Xiangrong Zhang, Tianyang Zhang, Xu Tang, Puhua Chen, and Licheng Jiao are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: xrzhang@mail.xidian.edu.cn).

Huiyu Zhou is with the School of Computing and Mathematical Sciences, University of Leicester, LE1 7RH Leicester, U.K.

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(a)

(b) Unbiased Teacher (c) Ours (d) Ground Truth

Fig. 1. Statement of motivation for this article. (a) Pseudo-labels change drastically at different training stages. (b)–(d) Comparison between the detection results of Unbiased Teacher and our approach, in contrast to the ground truth. Unbiased Teacher is more susceptible to confusion among similar categories, whereas our method effectively mitigates this problem.

we observed that classification errors in pseudo-labels with sufficient accurate localization mainly originate from confusion among similar categories [as shown in Fig. 1(b)]. We believe that the noise can be largely attributed to the poor discrimination of inter-class features learned by SSOD. While SSOD performs well on a few labeled data owing to deep learning's inherent powerful data fitting capability, the diversity of geospatial objects in RSIs hinders SSOD from capturing features with sufficient intra-class distinctiveness. This vulnerability makes SSOD more susceptible to being confused by categories with similar context or appearance. Existing methods treat all classes equally, which makes it more difficult for SSOD to distinguish these confusing categories.

To address the above challenges, we propose a novel temporal-feedback self-training (TST) framework for semi-supervised RSOD, consisting of two key components: temporal consistency-based pseudo-labels certainty estimation (TCE) and temporal self-feedback feature refinement (TSF). TCE measures the quality of pseudo-labels by combining their classification and localization stability over time series. Prior studies [18], [19] have demonstrated that deep learning first learns simple patterns before fitting noise data. Therefore, we consider pseudo-labels that exhibit smaller position and confidence fluctuations within a time window to be more reliable. Hence, we propose the TCE and assign the obtained certainty estimation of the unlabeled data as their dynamic weights in the loss function to reduce the influence of pseudo-label noise. Unlike these methods [16], [20], [21], [22], the proposed TCE bypasses the way to elaborately design confidence threshold since the complex distribution of noisy pseudo-labels is difficult to filter with certain thresholds [23], [24], [25]. Moreover, due to the limited diversity of labeled data, the model inherently lacks the ability to distinguish between similar categories. Therefore, we turn our focus to the source to improve the feature representation of easily confused categories by proposing the TSF. During SSOD learning, TSF dynamically identifies its own confusing categories as feedback for feature refinement, specifically improving the inter-class feature discrimination of these categories. TSF improves the feature representation of the data at the source, facilitating the creation of higher-quality pseudo-labels. TCE reduces the influence of pseudo-label noise during model learning. Both components drive the progress of self-training-based SSOD to achieve more stable and effective learning.

Our main contributions can be summarized as follows.

1) We propose a novel TST framework that effectively mitigates two key challenges in self-training-based semi-supervised RSOD methods: the noise of pseudo-labels caused by the instability of the model during the training process and insufficient discrimination of inter-class features learned by the model.
2) We propose a novel TCE based on the stability of pseudo-label classification and localization over time series to assess pseudo-label quality, mitigating the pseudo-label noise problem caused by model instability.
3) We propose a novel TSF to improve the inter-class feature discrimination learned by SSOD by dynamically detecting its own easily confused categories as feedback for feature refinement during training, thereby improving the quality of pseudo-labels.
4) We conducted extensive experiments on two authoritative RSOD datasets, and our method achieved state-of-the-art results, demonstrating its effectiveness.

II. Related Work

A. Remote Sensing Object Detection

Inspired by methods in natural image processing, mainstream RSOD approaches can be categorized into two-stage and single-stage. The former comprises rough region extraction followed by precise bounding box classification and localization, while the latter directly performs object classification and localization. Due to considerable scale variations of objects in RSIs, researchers have explored constructing multiscale feature representations to enhance detector performance [26]. This exploration mainly involves three aspects: multiscale feature integration [27], [28], [29], pyramid feature hierarchies [30], and feature pyramid networks [31], [32], [33]. However, objects in RSIs typically exhibit arbitrary orientations, making traditional horizontal bounding box representations insufficient for accurately localizing rotated objects. Consequently, subsequent researchers have predominantly focused on the design of algorithms for detecting rotated objects. Researchers have experimented with various representation methods, including five-parameter representation [34], [35], [36], eight-parameter representation [37], [38], [39], angle-based classification representation [40], and Gaussian distribution representation [41]. Nevertheless, these methods are all based on fully supervised learning and rely heavily on annotated data. This article focuses on semi-supervised RSOD, aiming to reduce the model's dependency on annotated data.

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B. Semi-Supervised Object Detection

The current mainstream SSOD approach is based on self-training paradigms [15], [42]. STAC [10] implements the combination of pseudo-labeling [5], [8] and consistent regularization [43], [44] methods. Instant-Teaching [12] instantaneously updates pseudo-labels online and introduces dual model co-rectify to improve the quality of pseudo-labels. Unbiased Teacher [9] follows the teacher–student network architecture [44] to update pseudo-labels online more stably. Soft Teacher [14] selects high-confidence pseudo labels for classification and reliable pseudo-boxes for regression by the predicted variance of box jitter. To mitigate the confirmation bias issue, Combating Noise [17] combines the classification scores of pseudo-boxes and IoU to quantify uncertainty. RPL [16] mitigates the amplified pseudo-label noise by combining the classification and localization scores of region proposals to dynamically adjust the category threshold of the pseudo-label. Some other works [23], [24], [45], [46] are devoted to designing pseudo-label assignment schemes to alleviate the noise of pseudo-labels. The most relevant works in SSOD to ours are ISMT [13] and VCL [47]. They both use historical information of pseudo-labels. Among them, ISMT believes that the differences in pseudo-labels generated at different training stages will hinder the convergence of SSOD. Therefore, it proposes using non-maximum suppression (NMS) to merge pseudo-labels from previous iterations and employs multiple detection heads with cross-supervision to provide complementary information. However, we argue that the indiscriminate use of all previous predictions as pseudo-labels by the ISMT cannot alleviate the noise in the pseudo-labels. Therefore, we exploit the temporal stability of pseudo-labels in classification and localization to alleviate the noise. VCL [47] aims at the utilization of ambiguous samples, by assigning the ambiguous samples whose categories change during the training process to an additional virtual category to optimize its features away from other categories. Although the above two works use historical temporal information of pseudo-labels, they are not explicitly designed for pseudo-label quality assessment. Our work mitigates the impact of noisy pseudo-labels by assessing their quality based on temporal stability in both classification and localization. While many other SSOD methods also explore different approaches to reducing pseudo-label noise, they do not address this issue from the perspective of temporal stability.

C. Semi-Supervised RSOD

In recent years, the remote sensing field has witnessed a notable surge in the adoption of SSL as a strategy for relieving the burden of data annotation [48], [49]. However, research focused on semi-supervised RSOD remains relatively scarce. The prospective consistent teacher (PCT) [50] introduced a novel approach for semi-supervised RSOD. This method utilizes box confidences to reweight dense proposals, leverages an EM-based pseudo-label filtering policy, and relies on the consistency between output boxes from both the teacher and the student to rerank student proposals. The SOOD model [51] designed rotation-aware adaptive weighting loss and global consistency loss to strengthen the instance and scene-level consistency between students' and teachers' predictions and achieve semi-supervised rotating object detection in RSIs. Unlike these previous works, we focus on the issues of model instability and insufficient inter-class feature discrimination that limits the performance of SSOD based on the self-training paradigm.

III. Method

A. Preliminary

Under the setting of SSOD, a set of labeled data $\boldsymbol{D}_l = \{\boldsymbol{x}_i^s, \boldsymbol{y}_i^s\}_{i=1}^{N_l}$ and unlabeled data $\boldsymbol{D}_u = \{\boldsymbol{x}_i^u\}_{i=1}^{N_u}$ are available during training. Here, $N_l \ll N_u$, and $\boldsymbol{x}$ and $\boldsymbol{y}$ represent image and ground truth annotations, respectively. We follow the mainstream self-training-based SSOD methods, which involve joint training of the labeled data $\boldsymbol{D}_l$ and unlabeled data $\boldsymbol{D}_u$ using a mutual learning paradigm with a teacher and student model [44]. The teacher model, updated through an exponential moving average (EMA) of the student model weights, makes predictions on weakly augmented versions of the unlabeled samples. Pseudo-labels are assigned to those samples whose predicted bounding box confidence exceeds a specified threshold σ (i.e., confidence threshold). The student model is trained with both the labeled data and the strongly augmented version of the unlabeled data. The final loss $\mathcal{L}$ is the weighted sum of the supervised loss $\mathcal{L}_s$ and unsupervised loss $\mathcal{L}_{\mathrm{u}}$ given as follows:

$$\mathcal{L} = \mathcal{L}_s + \lambda_u \mathcal{L}_{\mathrm{u}} \tag{1}$$

$$\mathcal{L}_s = \sum_i \mathcal{L}_{\mathrm{cls}}\left(\boldsymbol{x}_i^s, \boldsymbol{y}_i^s\right) + \mathcal{L}_{\mathrm{reg}}\left(\boldsymbol{x}_i^s, \boldsymbol{y}_i^s\right)$$

$$\mathcal{L}_u = \sum_i \mathcal{L}_{\mathrm{cls}}\left(\boldsymbol{x}_i^u, \hat{\boldsymbol{y}}_i^u\right) + \mathcal{L}_{\mathrm{reg}}\left(\boldsymbol{x}_i^u, \hat{\boldsymbol{y}}_i^u\right) \tag{2}$$

where $\mathcal{L}_{\mathrm{cls}}$ is the classification loss function, $\mathcal{L}_{\mathrm{reg}}$ is the regression loss function, and $\hat{\boldsymbol{y}}$ is the pseudo-labels of the unlabeled data.

As illustrated in Fig. 2, our proposed TST framework consists of two key components: TCE and TSF. In the figure, $p_1^{t_1}, p_1^{t_2}, p_1^{t_3}, and p_1^{t_4}$ $b_1^{t_1}, \ldots, b_1^{t_4}$ represent the class probabilities and bounding box positions for object 1 across time steps t_1 to t_4, respectively. Similarly, $p_2^{t_1}, p_2^{t_2}, p_2^{t_3}, p_2^{t_4}$ and $b_2^{t_1}, \ldots, b_2^{t_4}$ represent the corresponding values for object 2, and so on. The TCE filters out noisy pseudo-boxes caused by model instability and weights the loss function by considering the temporal stability of the remaining boxes in both classification and localization as a quality assessment. In addition, we propose the TSF to improve feature discrimination between confusable categories. In our training process, we introduce a feature memory bank to store the features of each category. The group of categories in which the model is more likely confused is obtained based on the degree of confusion of unlabeled data on the features of each category. We propose to use the spirit of contrast learning to improve the feature representation of the model by proposing the self-confusion contrast loss $\mathcal{L}_{sc}$. Thus, the overall loss function for our method is given as follows:

$$\mathcal{L} = \mathcal{L}_s + \lambda_u \mathcal{L}_u + \lambda_{sc} \mathcal{L}_{sc}. \tag{3}$$

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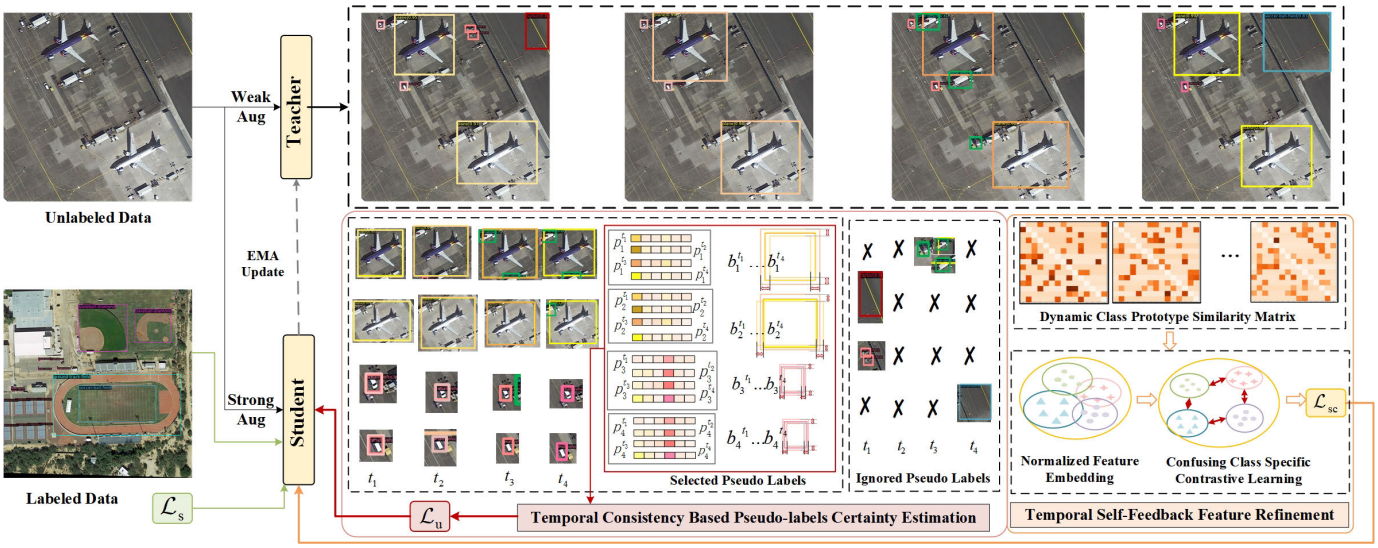


Fig. 2. Illustration of our proposed method. Our proposed TST framework consists of two key components: TCE and TSF. TCE employs the temporal stability of pseudo-labels in classification and localization to weight unlabeled data, mitigating the impact of pseudo-label noise. Meanwhile, TSF identifies easily confused categories during model training, refines their feature representations, and improves inter-class discrimination to generate more accurate pseudo-labels.

Algorithm 1 Temporal Consistency-Based Pseudo-Labels Certainty Estimation (TCE)

Require: Unlabeled data predictions $\boldsymbol{y}_i$, Time window length τ, IoU threshold $\mu = 0.5$

Ensure: Certainty estimation w_k^T and γ_k^T for each pseudo-box

```
Record pseudo-labels within time window [T − τ, T]
for each image do
    for each pseudo-box b_k^T at current time step T do
        Calculate the IoU between the b_k^T and each previous pseudo-box br_k^t for t ∈ [T − τ, T − 1]
        Select the predecessor br_k^t with the highest IoU that exceeds the threshold μ
        if No matching predecessor found then
            Mark as noisy box
        else
            Calculate position certainty u_k^T by averaging IoU with all predecessors
                Calculate the confidence certainty s_k^T by averaging the predicted probabilities with all predecessors.
            Compute combined certainty w_k^T using (4)
            Calculate boundary fluctuation γ_k^T using (6)
        end if
    end for
    Compute classification loss using (5)
    Compute regression loss using (8)
end for
Update model parameters
```

In Sections III-B and III-C, we provide a detailed explanation of TCE and TSF.

B. Temporal Consistency-Based Pseudo-Labels Certainty Estimation

SSOD based on the self-training paradigm requires generating pseudo-labels for unlabeled data while the model is being trained. However, the model's stability during the training is a concern. Particularly, the presence of complex backgrounds in RSIs intensifies the fluctuations in pseudo-label predictions due to model instability. The generated pseudo-label noise in turn misleads the learning of the model. Deep neural networks first learn simple patterns before overfitting noisy labels [19], therefore we tend to believe that pseudo-labels fluctuating less over time are cleaner. With this motivation, we propose the TCE, aiming to obtain a quality estimation of pseudo-labels by considering the stability of classification and localization on time series. The detailed steps of the TCE algorithm are outlined in Algorithm III-A.

To achieve our objective, it is necessary to record the pseudo-labels $\boldsymbol{y}_i$ for each image i within a time window. To reduce costs, we save the predictions of unlabeled data every 2000 iterations. The length of the time window is set to τ, which is a tunable parameter. Within the time window, historical predictions of τ times of unlabeled data are preserved, each considered as a time step t. Note that the first $\tau - 1$ times within the time window are the recent historical pseudo-labels every 2000 iterations, while the predictions at the current iteration T constitute the final time step of the time window. Thus, the time window can be represented as $[T - \tau, T]$. The pseudo-labels consist of (p_k^t, b_k^t), where b_k^t represents the top-left and bottom-right vertex coordinates of pseudo-box k, p_k^t represents its corresponding category probability distribution, and N_i^t is the number of pseudo-boxes for image i at time step t.

To evaluate the temporal stability of each pseudo-box, it is necessary to identify its state (e.g., position and category) at previous time steps. We compute the Intersection over Union (IoU) between the current pseudo-box b_k^T and each pseudo-box at previous time steps t, selecting the predecessor br_k^t with the highest IoU that exceeds a threshold μ (set to 0.5). This indicates that b_k^T has evolved from br_k^t at time step t. The IoU between the current pseudo-box b_k^T and its predecessor br_k^t is denoted as v_k^t, which reflects the degree of position change

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of b_k^T at time step t. However, if the current pseudo-box b_k^T has no matching predecessor, we consider this pseudo-box as noise generated by model instability and assign it to the noise set M_i^T. In such cases, the noisy pseudo-box is ignored when calculating the loss function. Otherwise, we assign the pseudo-box b_k^T to the certain boxes set B_i^T and calculate the location certainty u_k^T and the certainty of the confidence s_k^T. We compute the average predicted probability $\hat{p}_k^T$ of the current pseudo-box b_k^T over all its predecessors br_k^t at previous time steps, with the confidence certainty s_k^T being the maximum value of $\hat{p}_k^T$. The position certainty u_k^T is obtained by averaging the IoU values between the current pseudo-box b_k^T and its predecessors br_k^t at all previous time steps. We obtain the certainty estimation w_k^T of the pseudo-box b_k^T by linearly combining the stability of the localization and confidence as follows:

$$w_k^T = \begin{cases} \alpha_1 s_k^T + \alpha_2 u_k^T, & b_k^T \in B_i^T \\ 0, & b_k^T \in M_i^T. \end{cases} \tag{4}$$

We simply set $\alpha_1 = \alpha_2 = 0.5$. The classification loss function for image $\boldsymbol{x}_i$ is defined as follows:

$$\mathcal{L}_{\text{cls}}\left(\boldsymbol{x}_i^s\right) = \frac{1}{N_i^T}\left(\sum_{k=1}^{N_i^T}\sum_{j=1}^{n_k} w_k^T l_{\text{cls}}\left(y_k, q_j^k\right) + \sum_{j=1}^{M_i^{\text{bg}}} l_{\text{cls}}\left(y_{\text{bg}}, q_j^{\text{bg}}\right)\right). \tag{5}$$

Here, y_k is the one-hot vector of $\hat{p}_k^T$, and q^k represents the positive candidate boxes output by the RPN that matches pseudo-box k. n^k is the number of positive samples corresponding to the pseudo-class label y_k. Similarly, y_{bg} is the one-hot vector of the background class, and q^{bg} indicates the negative candidate boxes. M_i^{bg} represents the number of negative candidate boxes in the image i.

For the calculation of the regression loss, we first compute the fluctuation amplitude γ_k^T of the pseudo-box b_k^T, which is obtained by averaging the relative fluctuation of the coordinates between the current pseudo-box b_k^T and its predecessors br_k^t at other time steps, as follows:

$$\gamma_k^T = \frac{1}{N_r^T}\sum_{t=T-\tau}^{T-1}\left|\frac{b_k^T - br_k^t}{\sigma_k^T}\right|_{\text{abs}} \tag{6}$$

where σ_k^T represents the area of the pseudo-box b_k^T, $|\cdot|_{\text{abs}}$ indicates the absolute value operation, and N_r^T denotes the total number of predecessors for the pseudo-box b_k^T. The weight of the pseudo-box b_k^T for regression losses is determined as follows:

$$\omega_k^T = \begin{cases} \dfrac{w_k^T\left(1-\gamma_k^T\right)}{\max_{k=1}^{N_i^T}\left(1-\gamma_k^T\right)}, & b_k^T \in B_i^T \\ 0, & b_k^T \in M_i^T. \end{cases} \tag{7}$$

The regression loss for image i is as follows:

$$\mathcal{L}_{\text{reg}}\left(\boldsymbol{x}_i^s\right) = \frac{1}{N_i^T}\left(\sum_{k=1}^{N_i^T}\sum_{j=1}^{n_k}\omega_k^T l_{\text{reg}}\left(b_k^T, r_j^k\right)\right) \tag{8}$$

where r_k is the positive candidate box corresponding to the pseudo-box b_k^T.

C. Temporal Self-Feedback Feature Refinement

The powerful feature representation capability of deep learning models heavily relies on the quantity of labeled data. However, the limited diversity of labeled data causes SSOD to fail to learn strong discriminative features, which makes it challenging to identify unlabeled samples that have not been seen before. During our experiments, we found that a large amount of pseudo-label noise in unlabeled samples comes from the confusion between visually and contextually similar categories. This indicates the insufficient inter-class discrimination learned by SSOD among these categories. Existing SSOD algorithms learn all categories indiscriminately, which is quite inappropriate for SSOD. We believe that more attention should be paid to these confusing categories. To address this issue, we propose the TSF that aims to improve the feature representation capability of SSOD from the source. TSF enables SSOD to detect its own easily confused categories during learning and use this information as feedback to adjust the feature representation of the model, thereby contributing to the generation of more accurate pseudo-labels.

We maintain a feature memory bank of length $M_u + M_s$ for each category c during SSOD training to store features. M_u and M_s represent the memory bank length for unlabeled and labeled data, respectively. We directly take the ROI features of the ground truth for labeled data. For unlabeled data, we use the ROI features of the pseudo-box with confidence larger than 0.9. We transform the ROI features of the category c of unlabeled data and calculate the mean to obtain the class prototype vector for class c. This vector is obtained by averaging the transformed features of all the samples in category c. Specifically, the transformation involves flattening and applying a linear mapping operation. Clustering the class prototypes is an intuitive way to identify easily confused categories, but it can be time-consuming during training. Therefore, we adopt an alternative approach to identify the confused categories, by selecting the k most similar class prototypes to form the group of confused categories. Initially, we calculate the similarity between the class prototypes to generate the similarity matrix S as follows:

$$S_{j,k} = \begin{cases} \dfrac{o_j^T o_k}{\left\|o_j\right\| \cdot \left\|o_k\right\|}, & j \neq k \\ 0, & j = k \end{cases} \tag{9}$$

where j and k represent the row and column indices of matrix S, respectively, with $j, k \in [1, C]$, and C denotes the total number of categories. Subsequently, the top k most similar categories are selected for each category c to form its confuse set E_c as follows:

$$E_c = \left\{j \mid S_{c,j} \text{ in topk } (S_c)\right\}. \tag{10}$$

We then use contrast learning, which is a handy tool to improve the model's ability to distinguish between easily confused categories. For the ROI feature z_j, we use an MLP layer with negligible cost to encode the ROI feature, followed by L2 normalization to obtain the contrast feature $\widetilde{z}_j$.

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Inspired by [52], the supervised contrast loss can be expressed as follows:

$$L_{z_j} = -\frac{1}{M_{y_j}-1}\sum_{k=1,k\neq j}^{M}\mathbb{I}\{y_j = y_k\} \cdot \log\frac{\exp(\tilde{z}_j\cdot\tilde{z}_k/\varepsilon)}{\sum_{r=1}^{M}\mathbb{I}_{r\neq j}\cdot\exp(\tilde{z}_j\cdot\tilde{z}_r/\varepsilon)}. \tag{11}$$

Here, M is the number of proposals in the same batch, y_j and y_k denote the class labels of candidate box j and k, and M_{y_j} denotes the number of candidate boxes with class label y_j. ε is the temperature coefficient set to 0.1. However, such contrast learning equally impacts all categories, and the effort may be diluted by categories that are less prone to confusion. Therefore, we propose the self-confusion contrast loss in the following form:

$$\mathcal{L}_{sc}(z_j) = -\frac{1}{\mathrm{M}_{y_j}-1}\sum_{k=1,k\neq j}^{\mathrm{M}}\mathbb{I}\{y_j = y_k\}\cdot l_{kj}$$
$$l_{kj} = \log\frac{\exp(\tilde{z}_j\cdot\tilde{z}_k/\varepsilon)}{\exp(\tilde{z}_j\cdot\tilde{z}_k/\varepsilon)+\sum_{r=1,r\in E_{y_j}}^{\mathrm{M}_j}\exp(\tilde{z}_j\cdot\tilde{z}_r/\varepsilon)}. \tag{12}$$

Considering that the model's predictions may be unreliable in the early stages of training, it is reasonable to apply the proposed contrast loss during the middle of the training process rather than at the beginning. Thus, the final contrast loss for each image is defined as follows:

$$\mathcal{L}_{sc}(\boldsymbol{x}_i^s) = \begin{cases} \dfrac{1}{M}\displaystyle\sum_{j=1}^{M} u_j\cdot L_{z_j}, & I < I_m \\ \dfrac{1}{M}\displaystyle\sum_{j=1}^{M} u_j\cdot \mathcal{L}_{sc}(z_j), & I \geq I_m \end{cases} \tag{13}$$

where u_j is the IoU score of the candidate box j with its ground truth bounding box, I denotes the current iteration of the model, and I_m denotes the number of warm-up iterations used for our proposed contrastive loss.

IV. Experiments

A. Datasets and Evaluation

1) DOTA: It is a widely acknowledged and extensively utilized dataset for RSOD [53]. It contains 2806 images, segregated into the DOTA-train set, DOTA-val set, and DOTA-test set, which consists of 1411, 458, and 937 images, respectively. These images exhibit a diverse range of sizes, spanning from 800 × 800 to 4k × 4k, collectively incorporating 188 282 instances. These images exhibit a diverse range of sizes, spanning from 800 × 800 to 4k × 4k pixels, collectively incorporating 188 282 instances. These instances belong to 15 object classes, that is, plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). Following the previous convention [50], [51], we randomly divide 1%, 5%, 10%, 20%, and 30% of the data from the DOTA-train set as the labeled data and the rest as the unlabeled data. Throughout all experiments conducted on the DOTA dataset, we consistently evaluate performance on the DOTA-val set.

2) DIOR: This is another publicly available and extensively employed dataset [54] for RSOD, which includes 23 463 images and 190 288 instances. It consists of three datasets: DIOR-train, DIOR-val, and DIOR-test, each hosting 5862, 5863, and 11 738 images, respectively. The size of these images is 800 × 800 pixels. The dataset is thoughtfully classified into 20 categories, including airplane (APL), airport (APR), baseball field (BF), BC, bridge (BR), chimney (CM), dam (DAM), expressway service areas (ESA), expressway toll station (ETS), golf field (GF), GTF, harbor (HB), overpass (OP), ship (SH), stadium (SD), ST, tennis court (TC), train station (TS), vehicle (VC), and windmill (WM). In our experiments, we refer to the setting of the VOC dataset in SSOD, using the DIOR-train set as labeled data and the DIOR-test set as unlabeled data for performance evaluation on the DIOR-val set.

3) Metrics: Following previous conventions for RSOD, we use standard mean average precision (mAP) as our evaluation metric.

B. Implementation Details

We follow previous SSOD approaches [9], [12], [12], [14] and employ the Faster R-CNN detector [55] with a ResNet-50-FPN [31], [56] backbone in our experiments to ensure fairness and correctness of comparisons. Data augmentation is consistent with the Unbiased Teacher [9], and model weights are initialized from models pretrained on ImageNet. We perform experiments on an NVIDIA 3090 using an SGD optimizer with a learning rate of 0.005, a momentum rate of 0.9, and a weight decay of 0.0001. Our main experiments set the batch size to 10 and select a confidence threshold σ of 0.7 following the Unbiased Teacher [9]. The weight for unlabeled data is λ_u of 1.0 and the contrast loss weight is λ_{sc} of 0.03. The length of the time window τ is 4, the value of the number of confusion categories k chosen is 4, and I_m is the 14th epoch. M_u and M_s are set to 60. During training, the memory bank dynamically stores 120 features per category, each as a 256D vector. The bank is updated at each step using a first-in, first-out (FIFO) queue, ensuring minimal computational overhead. For all experiments, we trained 28 epochs. Our code is available at https://github.com/ZxqGoAhead/SemiRSOD-TST

C. Comparison to the State-of-the-Art

1) Overall Performance Comparison: We compare our approach with the state-of-the-art SSOD methods on both the DOTA and DIOR datasets. The results are shown in Table I. The table illustrates that our method achieved mAP scores of 41.7%, 61.8%, 63.7%, 68.7%, and 70.1% on the 1%, 5%, 10%, 20%, and 30% protocols of the DOTA dataset, respectively. Additionally, our method achieved an 82.8% mAP score on the DIOR dataset. Our method outperformed recent state-of-the-art general SSOD methods, achieving the highest mAP

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TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS

Method	DOTA					DIOR
	1%	5%	10%	20%	30%	
Supervised	28.6	50.1	52.6	54.2	56.4	72.0
Unbiased Teacher [9]	35.4	53.1	56.6	58.7	62.0	75.9
Soft Teacher [14]	37.1	57.6	58.1	61.7	66.3	79.7
Dense Teacher [24]	40.0	56.6	60.1	64.0	65.8	79.6
Unbiased Teacher V2 [15]	40.5	57.7	60.3	64.6	65.4	80.5
Consistent Teacher [57]	38.5	58.7	60.6	62.4	63.7	78.1
PCT [50]	40.9	60.2	61.7	-	-	-
TST(Ours)	**41.7**	**61.8**	**63.7**	**68.7**	**70.1**	**82.8**

scores on two authoritative RSOD datasets. Compared with natural images, RSIs exhibit more prominent complexity and variability, exacerbating pseudo-label noise in semi-supervised RSOD. As evident from Table I, general SSOD methods are inadequate in addressing this issue effectively.

Our method yields remarkable improvements in mAP scores compared to fully supervised models trained using only labeled data. Notably, our method exhibits improvements of 13.1%, 11.7%, 11.1%, 14.5%, and 13.7% on the 1%, 5%, 10%, 20%, and 30% protocols of DOTA, respectively, along with a 10.8% enhancement in mAP on DIOR. These results strongly demonstrate the effectiveness of our method in leveraging unlabeled RSIs. PCT is the state-of-the-art method in recent semi-supervised RSOD. Our method outperformed PCT by margins of 0.8%, 1.6%, and 2% in mAP on the 1%, 5%, and 10% protocols of DOTA, respectively. Despite PCT leveraging the consistency of unsupervised branches and adapting pseudo-label thresholds, it fails to adequately address the pseudo-label noise from model instability and struggles to differentiate easily confounded categories, thereby limiting its effectiveness. Conversely, our method takes full advantage of the prefitting properties of deep networks on simple, clean data, while also addressing pseudo-label noise in classification and localization. Combined with our efficient pseudo-label quality evaluation method, our method can effectively handle the complexity and variability of RSIs.

2) Comparison of Different Categories: To provide a comprehensive performance comparison, we present the scores of comparative methods for each category in both the DOTA and DIOR datasets, as shown in Tables II and III. Unbiased Teacher serves as the baseline for our method. In both tables, the red number indicates the highest score for the corresponding category, and the blue number indicates the second-highest score.

Table II presents the performance comparison of different categories across all DOTA dataset protocols. On the 1% protocol, our method exhibits improvements over the Unbiased Teacher in the PL, BD, BR, GTF, LV, TC, ST, RA, and SP categories. However, with only 1% of labeled data, TST may lack robustness and fall behind Unbiased Teacher in the SV, SH, BC, and SBF. Nevertheless, TST ultimately achieves a 6.3% higher mAP than Unbiased Teacher. On the 5% protocol, TST improves the AP of 12 categories, with only slight lags behind Unbiased Teacher in the LV and BC, ultimately achieving an mAP 8.7% higher than Unbiased Teacher. On the 10% protocol, TST almost improves the AP of 13 categories compared to Unbiased Teacher, ultimately achieving an mAP of 7.1% higher than Unbiased Teacher, with only slight lags behind in the LV, BC, and HC categories.

On the 20% protocol, TST improves the AP of almost all categories, with only a slight lag behind Unbiased Teacher in the SV. Particularly, the GTF, SBF, RA, SP, and HC witness significant improvements, with AP increases of 23.7%, 17.3%, 13%, 35.1%, and 14.6%, respectively, leading to a 10% mAP increase for TST. On the 30% protocol, TST boosts the AP of nearly all categories, with only a slight lag in the LV category, resulting in an mAP of 8.1% higher than Unbiased Teacher. Significant improvements are observed, particularly in the GTF, RA, SP, and HC. The table reveals that the categories in which TST surpasses Unbiased Teacher are not consistently fixed. Moreover, the improvement is relatively minor for categories where Unbiased Teacher performs well, but more significant for categories where Unbiased Teacher performs poorly. We attribute this phenomenon to the dynamic nature of the TST component in our method, which consistently identifies categories prone to self-confusion, enhances discrimination between them, mitigates weaknesses, and simultaneously preserves strengths.

Table III presents the AP performance of different methods across various categories on the DIOR dataset. TST exhibits improvements across all categories compared to Unbiased Teacher. Particularly noteworthy improvements are observed in the APR, BR, ETS, GF, HB, and TS, with AP increases of 9.5%, 10.7%, 17.7%, 12.8%, 12.7%, and 19%, respectively, resulting in a final mAP increase of 6.9%. It can be observed from the table that our method exhibits substantial improvements in categories where Unbiased Teacher performs poorly, while still showing notable enhancements in categories where Unbiased Teacher performs well. This result reflects the robustness and effectiveness of our approach.

3) Visual Prediction Comparison: Figs. 3 and 4 present the prediction visualizations of recent state-of-the-art methods and our proposed method under the 1% and 30% protocols on the DOTA dataset, respectively. In Fig. 3, Unbiased Teacher missed several LVs and confused them with ships and helicopters. It also tended to confuse ships with harbors and some background objects when detecting harbors. Other advanced methods similarly exhibited significant missed detections. Due to the extremely limited labeled data in the 1% protocol, Unbiased Teacher suffered from a severe lack of category discrimination capability, a shortcoming effectively addressed by our method. In Fig. 4, Unbiased Teacher struggled to distinguish between airplanes and helicopters and confused the categories of GTF and BD. Despite the larger amount of labeled data in the 30% protocol, Unbiased Teacher still showed class confusion. Our method, with the proposed TSF component, significantly improved the model's ability to differentiate these challenging categories. Additionally, both other state-of-the-art methods and Unbiased Teacher exhibit inadequate localization capabilities in detecting harbors and soccer fields, while our proposed method effectively mitigated the localization noise in pseudo-labels.

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TABLE II

COMPARISON OF PERFORMANCE FOR SPECIFIC CATEGORIES ON THE DOTA DATASET, WITH UNBIASED TEACHER AS THE BASELINE FOR OUR METHOD

Percentage	Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
1%	Supervised	71.3	4.3	10.8	4.8	45.6	40.5	53.9	67.0	2.6	47.4	15.6	10.5	32.9	11.4	11.0	28.6
	Soft Teacher	87.1	35.9	22.5	26.3	40.5	14.9	32.9	90.8	16.1	50.6	50.3	26.6	38.1	17.1	7.5	37.1
	Unbiased Teacher V2	87.4	44.8	29.4	23.8	28.8	20.4	32.3	92.0	16.3	63.8	47.1	47.4	38.4	29.1	7.2	40.5
	Dense Teacher	84.2	33.3	19.4	32.6	14.3	52.2	55.4	87.3	4.8	60.1	33.9	27.2	41.2	44.7	8.2	40.0
	Consistent-Teacher	83.6	18.7	18.4	17.7	36.1	51.2	60.6	89.8	2.2	54.1	19.7	37.6	34.0	44.6	9.0	38.5
	Unbiased Teacher	81.2	9.1	13.6	10.9	47.2	56.6	73.5	83.0	5.2	51.7	25.4	8.9	47.6	9.8	6.7	35.4
	TST(Ours)	83.5	25.9	20.0	23.6	38.3	58.5	65.5	87.5	3.5	54.0	15.0	47.4	47.7	45.6	9.3	**41.7**
5%	Supervised	81.0	45.6	23.3	22.6	53.1	72.3	82.1	88.7	55.1	57.3	34.6	38.1	65.9	10.2	21.5	50.1
	Soft Teacher	82.4	61.1	35.4	43.8	52.1	71.8	73.2	91.7	48.3	64.8	45.8	44.2	66.1	43.2	33.3	57.6
	Unbiased Teacher V2	85.5	66.3	38.4	43.8	40.2	68.9	54.0	93.0	50.1	60.9	54.7	52.4	63.8	47.3	45.5	57.7
	Dense Teacher	83.6	64.6	26.7	34.3	30.0	75.4	72.3	92.7	52.0	60.8	54.4	52.8	72.0	48.2	29.3	56.6
	Consistent Teacher	85.3	70.6	38.2	58.9	35.8	75.8	69.1	92.8	53.3	56.2	52.5	53.6	70.8	41.1	26.5	58.7
	Unbiased Teacher	83.7	53.8	26.0	30.9	53.3	74.0	83.4	90.1	51.3	55.3	41.3	43.0	70.0	15.4	25.5	53.1
	TST(Ours)	86.8	64.7	34.4	55.7	54.2	72.7	83.4	93.1	50.9	63.1	55.1	59.2	73.2	48.9	32.0	**61.8**
10%	Supervised	82.9	54.4	23.4	26.2	55.4	72.5	84.0	89.1	59.2	58.2	37.1	37.4	70.2	13.2	26.3	52.6
	Soft Teacher	88.8	73.7	30.5	30.6	44.6	75.7	56.3	93.3	61.9	57.2	51.0	52.0	75.6	51.7	28.4	58.1
	Unbiased Teacher V2	87.2	73.4	40.8	53.8	38.8	69.7	48.8	93.3	63.7	62.4	58.4	57.6	73.2	53.7	30.1	60.3
	Dense Teacher	87.6	59.7	32.4	34.4	45.8	79.0	66.9	93.0	66.6	71.9	53.7	60.2	70.3	49.9	30.6	60.1
	Consistent Teacher	85.3	71.1	39.3	60.5	44.4	79.1	75.9	92.5	56.1	57.6	55.5	55.5	72.9	41.5	21.4	60.6
	Unbiased Teacher	85.4	58.1	27.8	34.0	54.0	75.6	84.8	91.2	62.4	58.4	45.9	46.6	72.7	19.5	33.2	56.6
	TST(Ours)	88.8	63.1	36.0	55.8	56.5	79.4	86.8	92.8	57.9	65.0	54.2	58.7	77.0	51.7	32.7	**63.7**
20%	Supervised	80.3	55.4	30.9	39.4	54.0	56.3	63.4	79.6	52.4	61.2	45.7	51.1	71.6	24.1	47.2	54.2
	Soft Teacher	89.2	75.9	39.1	47.0	45.0	76.0	58.5	93.7	67.3	56.7	48.3	53.3	78.3	53.6	43.4	61.7
	Unbiased Teacher V2	89.1	74.9	43.8	64.7	38.6	70.7	49.7	94.1	64.7	63.5	63.3	60.4	78.4	57.2	56.2	64.6
	Dense Teacher	91.3	65.9	36.7	37.7	48.3	81.0	76.3	93.5	63.9	72.3	54.0	57.6	74.5	56.7	50.6	64.0
	Consistent Teacher	85.0	72.1	39.2	64.4	47.1	78.7	79.3	92.8	56.3	61.5	53.1	52.8	77.3	51.3	25.5	62.4
	Unbiased Teacher	85.9	58.2	30.4	38.4	63.5	78.6	86.6	91.5	55.8	61.5	44.9	46.4	75.4	20.5	42.7	58.7
	TST(Ours)	90.9	70.2	39.4	62.1	60.2	80.0	86.9	92.6	63.7	65.2	62.2	59.4	78.5	55.6	63.4	**68.7**
30%	Supervised	82.5	61.1	34.5	40.0	57.1	58.4	63.8	80.9	52.6	62.5	52.1	55.2	73.5	26.4	45.4	56.4
	Soft Teacher	89.4	78.5	42.5	73.4	44.5	77.0	57.1	93.4	71.4	53.3	64.8	60.1	79.5	57.5	51.7	66.3
	Unbiased Teacher V2	89.4	75.2	46.2	63.8	39.1	76.1	50.4	93.4	62.2	65.9	65.5	62.4	80.2	54.8	56.3	65.4
	Dense Teacher	90.9	66.7	40.1	39.8	57.5	82.6	89.1	93.7	63.2	74.9	54.9	60.3	76.5	58.8	37.5	65.8
	Consistent Teacher	86.8	70.7	43.7	59.4	44.5	78.9	78.9	93.0	57.4	62.4	54.9	52.5	77.0	53.9	41.0	63.7
	Unbiased Teacher	87.8	63.8	37.2	46.7	62.9	82.7	88.0	92.1	55.7	61.6	50.9	50.9	77.6	28.6	43.7	62.0
	TST(Ours)	91.6	69.2	41.3	61.8	63.7	81.9	88.9	93.9	64.8	67.3	63.4	65.1	80.3	61.3	57.2	**70.1**

TABLE III

COMPARISON OF PERFORMANCE FOR SPECIFIC CATEGORIES ON THE DIOR DATASET, WITH UNBIASED TEACHER AS THE BASELINE FOR OUR METHOD

Method	APL	APR	BF	BC	BR	CM	DM	ESA	ETS	GF	GTF	HB	OP	SH	SD	ST	TC	TS	VC	WM	mAP
Supervised	91.0	65.8	92.4	81.8	38.8	90.9	55.7	62.6	57.9	76.9	79.5	50.2	60.0	78.7	90.6	81.1	89.2	49.0	71.2	77.6	72.0
Soft Teacher	92.1	84.7	96.7	83.3	50.8	90.4	69.2	70.4	77.2	87.0	91.4	55.1	68.4	80.2	95.8	82.5	96.9	68.1	75.8	81.5	79.9
Unbiased Teacher V2	93.7	86.5	93.5	88.0	51.1	93.6	72.6	73.7	78.2	84.1	87.7	58.4	67.5	75.2	95.8	81.8	96.3	66.1	81.3	85.9	80.5
Dense Teacher	91.3	83.5	93.9	87.2	53.2	90.0	65.2	77.2	75.8	83.5	86.6	60.0	68.2	79.7	92.9	83.9	96.0	69.4	77.1	83.0	79.6
Consistent Teacher	92.6	62.2	93.8	86.8	54.0	89.3	73.4	69.2	68.3	74.0	87.4	51.9	69.0	89.1	95.4	82.8	91.6	70.1	76.0	86.0	78.1
Unbiased Teacher	90.5	77.7	92.2	83.7	48.8	89.5	62.8	67.7	59.4	76.1	83.0	61.2	66.0	80.0	96.7	82.7	93.5	50.5	71.3	84.1	75.9
TST(Ours)	92.1	87.2	96.3	88.7	58.7	93.2	66.0	74.9	77.1	88.9	88.6	73.9	71.3	85.5	97.2	84.7	97.3	69.5	77.1	86.9	**82.8**

Fig. 5 shows the prediction results of various methods on the DIOR dataset, with different colors indicating different classes. Many methods, including Unbiased Teacher, Soft Teacher, Dense Teacher, and Unbiased Teacher V2, missed many densely packed ships and failed to identify harbors, as also observed with Consistent Teacher and Soft Teacher. In the second row, Unbiased Teacher and most other advanced methods confuse STs with ships and harbors. In the third row, Unbiased Teacher also exhibits confusion between background objects and categories such as GFs, ships, and dams. Our method effectively avoids background interference by estimating the uncertainty of pseudo-labels in both classification and localization.

D. Ablation Study

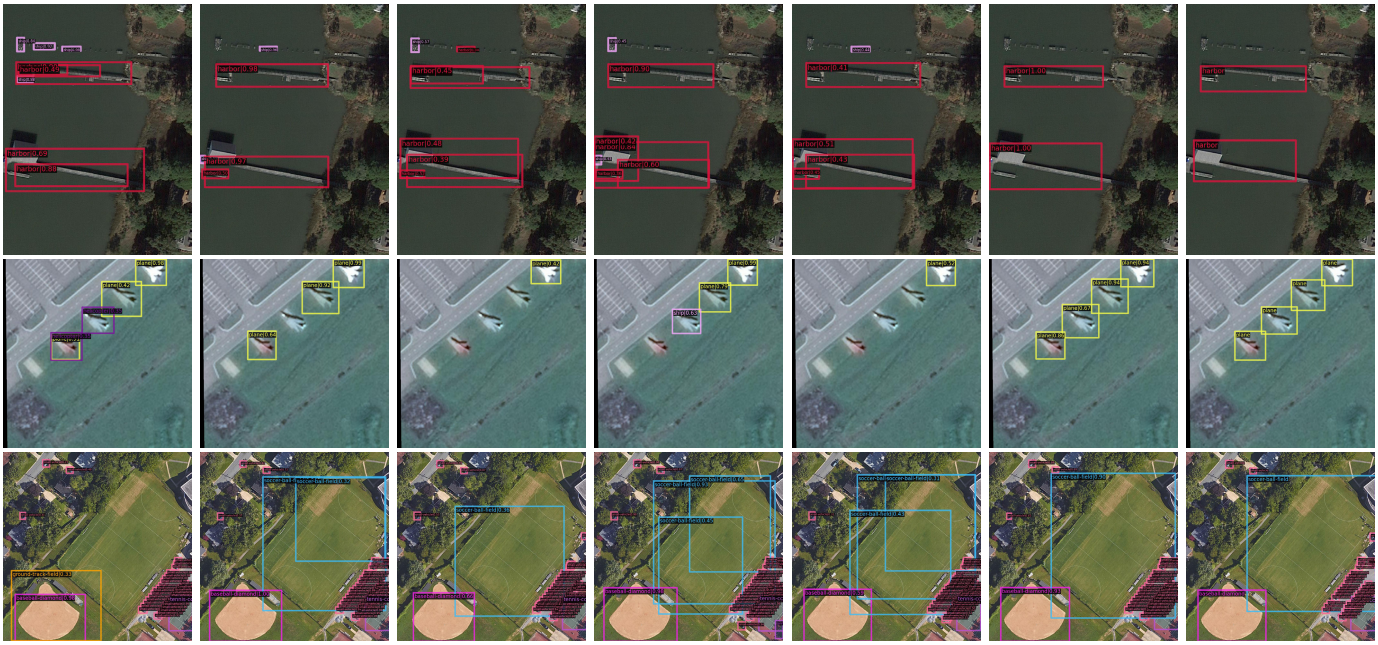
1) Effectiveness of TSF: Due to the lack of labeled data, SSOD struggles to learn sufficiently discriminative features, especially for categories with similar appearance or context. To address this issue, we propose the TSF that dynamically detects confusing categories during model training and improves the feature representation for those categories accordingly. As shown in Table IV, compared with the baseline, TSF improves mAP from 75.9% to 79.8% on the DIOR datasets, with an increase of 4.8%. For supervised contrastive learning (SupContrast) [52], mAP only improves by 0.7%. As shown in Fig. 6, TSF significantly increases the inter-class separability of the model's easily confused categories (i.e., basketball court, tennis court, ground track field, expressway toll station), further demonstrating the effectiveness of our proposed method. Fig. 7(b) illustrates the average precision (AP) of pseudo-labels on the DIOR datasets, indicating a significant improvement in category recognition accuracy achieved by TSF. In Fig. 8, we also show the comparison results of the category confusion matrix predicted by the proposed method

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Unbiased Teacher | Soft Teacher | Dense Teacher | Unbiased Teacher V2 | Consistent Teacher | Ours | Ground Truth

Fig. 3. Comparison of prediction visualization results of different methods from the DOTA test images on the 1% protocol.



Unbiased Teacher | Soft Teacher | Dense Teacher | Unbiased Teacher V2 | Consistent Teacher | Ours | Ground Truth

Fig. 4. Comparison of prediction visualization results of different methods from the DOTA test images on the 30% protocol.

on the DOTA test dataset without and with the TSF component. It can be seen from the figure that the TSF effectively reduces the confusion between similar categories, such as helicopters and planes, BDs, GTFs, and SBFs, etc. Moreover, we present the T-SNE feature visualization results for baseline, baseline + SupContrast, and baseline + TSF.

2) Effectiveness of TCE: The TCE aims to evaluate the quality of pseudo-labels by leveraging the stability of their confidence and position in the temporal sequence. We consider that pseudo-labels with less fluctuation are of higher quality and should play a more important role in the model learning process, thereby mitigating the influence of noisy pseudo-labels. To demonstrate the effectiveness of TCE, we conducted

TABLE IV
PERFORMANCE OF DIFFERENT COMPONENTS ON DIOR DATASETS

Methods	mAP
Baseline	75.9
+SupContrast	76.6
+TCE	81.7
+TSF	79.8
+TCE+TSF	82.8

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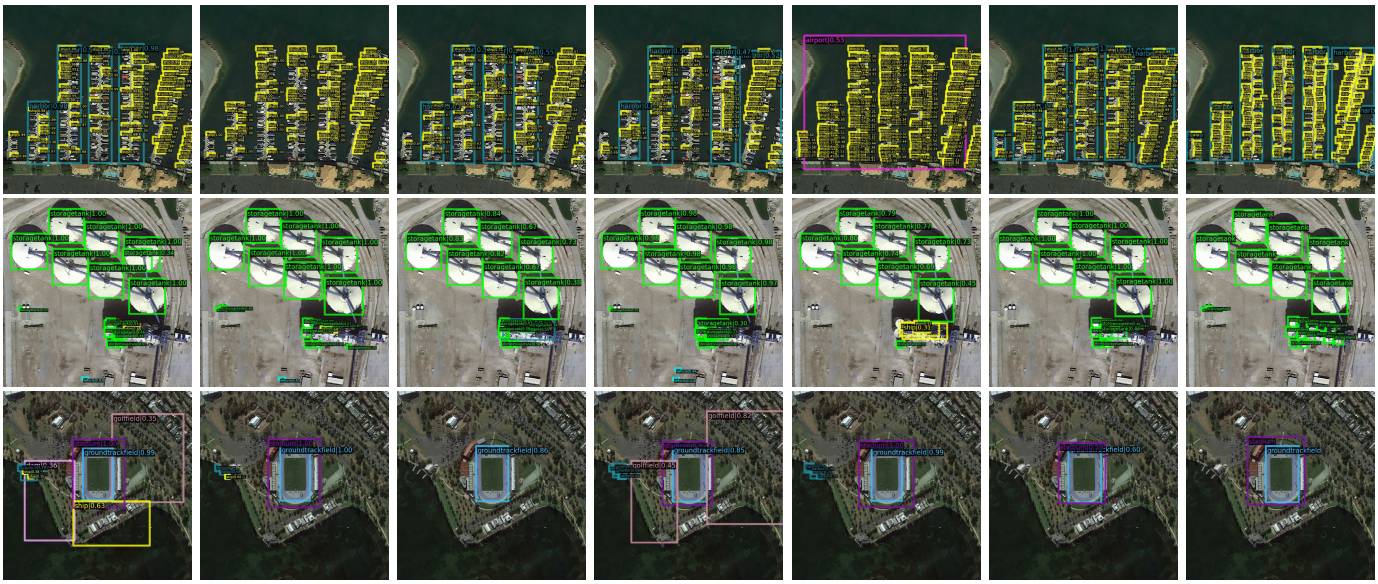


Fig. 5. Comparison of prediction visualization results of different methods from the test images of DIOR.

TABLE V
COMPARISON OF THE PERFORMANCE OF DIFFERENT τ IN THE TCE ON DIOR DATASETS

τ	mAP
Baseline	75.9
2	78.8
3	80.9
4	81.7
5	81.2

TABLE VI
COMPARISON OF THE PERFORMANCE OF DIFFERENT k IN THE TSF ON DIOR DATASETS

k	mAP
Baseline	75.9
2	77.4
3	78.5
4	79.8
5	79.2
20(Supcontrast)	76.6

ablation experiments on the DIOR datasets. As shown in Table IV, TCE significantly improved the baseline, with mAP increasing from 75.9% to 81.7%, representing increases of 5.8%. Moreover, as shown in Fig. 7(a), the quality of pseudo-labels is greatly improved with the use of TCE, providing further evidence of its effectiveness.

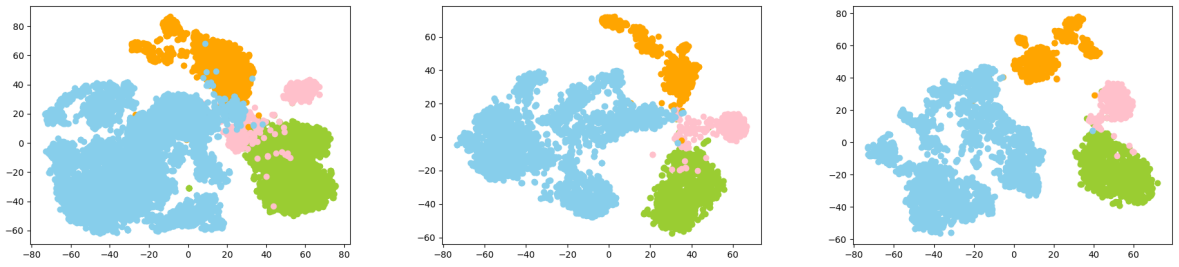
3) Different τ for TCE: We varied the value of τ to analyze the effect of the TCE. τ represents the length of the time series used to compute the stability of pseudo-labeling in TCE. As shown in Table V, on the DIOR dataset, TCE achieved the greatest improvement over baseline when τ is set to 4, with increases of 5.8% in mAP. In contrast, when τ is set too small, such as 2, the increase is only 2.9%. These results suggest that, if the length of the temporal window is too short, TCE may not effectively identify fluctuations in pseudo-labels, resulting in inaccurate quality evaluation. As τ increases, the effectiveness of TCE also improves. When τ is set to 4 and 5, the performance improvement was similar, with increases of 5.8% and 5.3% in mAP, suggesting that values in this range are more suitable.

4) Different k for TSF: We varied the parameter k to evaluate the effectiveness of TSF on the DIOR dataset. This parameter controls the number of confusable categories considered in TSF and determines the number of negative samples used for each category. As shown in Table VI, when $k = 2$, TSF provides only limited improvement over the baseline, with an increase of only 1.5% in mAP. This indicates that when k is too small, the number of negative samples for confusable categories is insufficient to effectively enhance the inter-class discriminability of the feature representation through contrastive learning. However, as k increases, the performance of TSF improves significantly. The maximum improvement is achieved with $k = 4$, where mAP is increased by 3.9%. When $k = 5$, the mAP is increased by 3.3%. However, as k becomes too large and reaches the number of categories (i.e., 20), performance significantly deteriorates and is equivalent to supervised contrastive learning [52]. This suggests that, when k is too large, the efforts of contrastive learning may be diluted by samples that the model easily distinguishes.

E. Extended Experiments

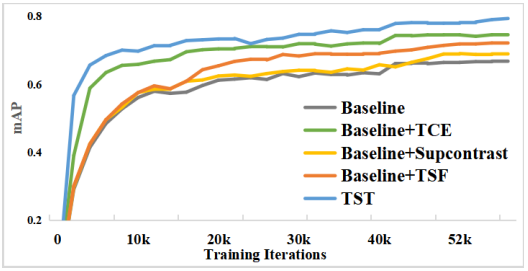
We evaluated the generalizability of our approach using the PASCAL-VOC [62] dataset for natural scene images. Following the established protocols of previous SSOD methods, we use the PASCAL-VOC07 train dataset as the labeled dataset and the PASCAL-VOC12 dataset as the unlabeled

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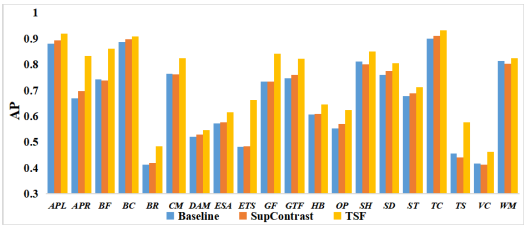


Baseline Baseline+SupContrast Baseline+TSF

Fig. 6. T-SNE features visualized by different methods in the DIOR dataset. Orange, blue, yellow-green, and pink correspond to the class GTF, TC, BC, and ETS, respectively.

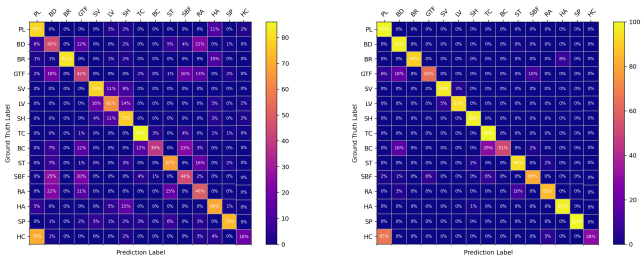


(a)



(b)

Fig. 7. Illustration of the effectiveness of the components of the proposed method on the DIOR dataset. (a) Comparison of pseudo-label mAP with and without TCE. (b) Comparison of AP for baseline, supervised contrastive learning, and proposed TSF pseudo-labels across various categories on the DIOR dataset.



(a) (b)

Fig. 8. Confusion matrices with and without the TSF component: (a) without TSF and (b) with TSF, under the DOTA 1% protocol.

TABLE VII

EXPERIMENTAL RESULTS ON PASCAL-VOC07 TEST DATASET FOR DIFFERENT METHODS

Methods	mAP
Supervised	72.63
CSD [58]	74.70
STAC [10]	77.45
Instant Teaching [12]	78.3
ISMT [13]	77.23
Humble Teacher [59]	80.94
Unbiased Teacher [9]	77.37
Combating Noise [17]	80.6
RPL [16]	79.0
DSL [25]	80.7
Unbiased Teacher V2 [15]	81.29
SED [60]	80.60
DDT [61]	82.4
Consistent-Teacher [57]	81.00
TST(Ours)	**84.1**



Unbiased Teacher Ours Ground Truth

Fig. 9. Comparison of the pseudo-label quantitative results of our method with the Unbiased Teacher on the PASCAL-VOC12 dataset.

training set, with performance evaluations conducted on the VOC07 test dataset. Experimental results are presented in Table VII, indicating that our method achieved an outstanding mAP of 84.1%, demonstrating its effectiveness in enhancing SSOD within the self-training paradigm and its generalizability in the relatively simple natural scenes. This success is attributed to our resolution of two critical issues previously neglected in SSOD: the instability of pseudo-labels and the insufficient discrimination of similar categories. Consequently, we developed two innovative components, TCE and TSF, to effectively address these challenges.

Fig. 9 provides a comparative analysis of pseudo-label generation between our method and the Unbiased Teacher on the PASCAL-VOC dataset, showing our method's superior ability to differentiate between similar categories such as cows,

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sheep, and horses, as well as its improved object localization accuracy. This demonstrates our method's capacity to diminish confusion among similar categories and reduce pseudo-label noise, thereby boosting model performance.

V. Conclusion

We propose the TST framework to address two challenges in self-training-based RSOD: pseudo-label noise and difficulty in distinguishing similar categories. TST includes two components: TCE and TSF. TCE assesses pseudo-label quality by evaluating its stability over time and adjusting loss function weights to reduce noise. TSF refines feature representation by detecting confusing categories and enhancing inter-class discrimination, improving pseudo-label quality. Experiments on the DOTA and DIOR datasets show that our method outperforms standard SSOD approaches, particularly for complex RSIs. Ablation studies confirm that both components improve RSOD performance. TST and TSF add minimal overhead during training, without impact on inference. We believe that our approach is highly scalable and can be applied to other SSL scenarios and noisy label tasks, such as semi-supervised semantic segmentation, domain-adaptive object detection, and social media text categorization. We plan to explore these applications in future work.

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Xiaoqian Zhu received the B.S. degree in intelligent science and technology from Xidian University, Xi'an, China, in 2019, where she is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education.

Her research interests include deep learning, semantic segmentation, object detection, semi-supervised learning, and remote sensing image understanding.



Xiangrong Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees in computer application technology from the School of Computer Science, Xidian University, Xi'an, China, in 1999 and 2003, respectively, and the Ph.D. degree in pattern recognition and intelligent system from the School of Electronic Engineering, Xidian University, in 2006.

She is currently a Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. Her research interests include pattern recognition, machine learning, and remote sensing image analysis and understanding.



Tianyang Zhang received the B.S. degree in intelligent science and technology from Xidian University, Xi'an, China, in 2018, and the Ph.D. degree in computer science and technology from Xidian University, in 2023.

He is currently a Post-Doctoral with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. His research interests include scene classification and object detection in remote sensing images.

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Xu Tang (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electronic circuit and system from Xidian University, Xi'an, China, in 2007, 2010, and 2017, respectively.

From 2015 to 2016, he was a Joint Ph.D. Student along with Prof. W. J. Emery with the University of Colorado at Boulder, Boulder, CO, USA. He is currently an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. His research interests include remote sensing image content-based retrieval and reranking, hyperspectral image processing, remote sensing scene classification, and object detection.



Puhua Chen (Senior Member, IEEE) received the B.S. degree in environmental engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2009, and the Ph.D. degree in circuit and system from Xidian University, Xi'an, China, in 2016.

She is currently a Lecturer with the School of Artificial Intelligence, Xidian University. Her research interests include machine learning, pattern recognition, and remote sensing image interpretation.



Huiyu Zhou received the B.Eng. degree in radio technology from the Huazhong University of Science and Technology, Wuhan, China, in 1990, the M.Sc. degree in biomedical engineering from the University of Dundee, Dundee, U.K., in 2002, and the Ph.D. degree in computer vision from Heriot-Watt University, Edinburgh, U.K., in 2006.

He is currently a Full Professor with the School of Computing and Mathematical Sciences, University of Leicester, Leicester, U.K. He has authored or co-authored over 380 peer-reviewed papers in the field.



Licheng Jiao (Fellow, IEEE) received the Ph.D. degree in signal, electronic circuits and systems from Xi'an Jiaotong University, Xi'an, China, in 1990.

From 1990 to 1991, he was a Post-Doctoral Fellow with the National Key Laboratory for Radar Signal Processing, Xidian University, Xi'an, where he is currently the Dean of the Electronic Engineering School and the Director of the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education. His research interests include signal and image processing, machine learning, natural computation, and intelligent information processing.

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